Apply definitions of decibels and decades

1. (vin/vout to dB) If an electronic component takes a 1.0v sine wave as input and output a 0.1v sine wave, how much attenuation would it have?

2. (vin/dB to vout) If you input a 2.4 V sin wave into a circuit that attenuated it by -32dB, what would the amplitude of the output waveform be?

3. (out/dB to vin) A circuit attenuates an input signal by -68 dB to produce a 0.5 V signal. What is the amplitude of the input signal?

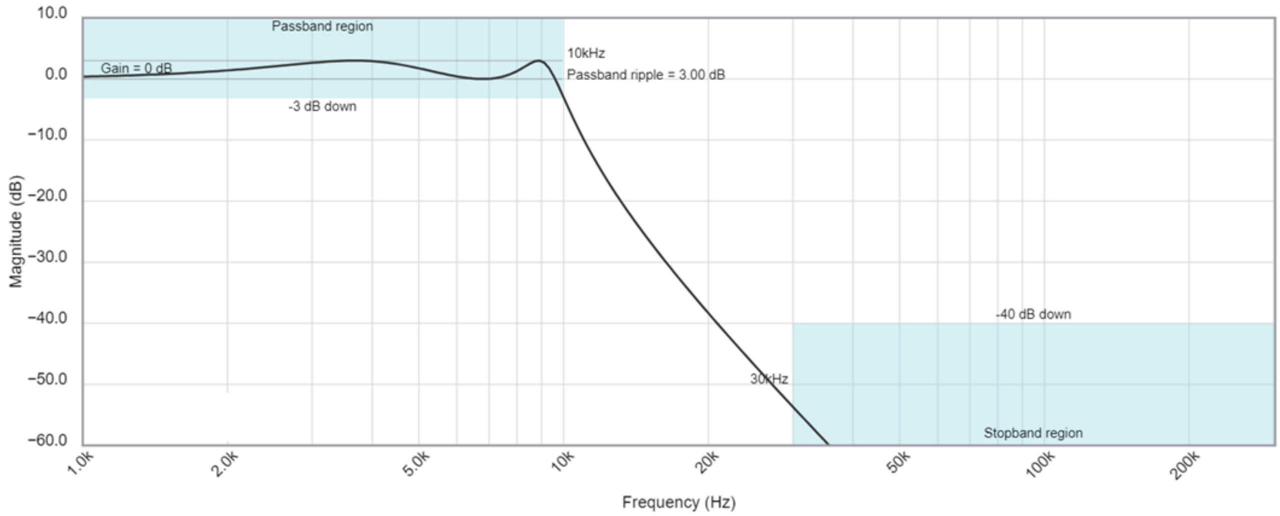
4. (fhi/flo to decades) How many decades separate 10kHz and 30kHz?

5. (fhi/decades to flo) What frequency is 1.5 decades below 300kHz?

6. (flo/decades to fhi) What frequency is 1.5 decades above 10kHz?

Interpret a Bode plot

The following Bode plot is from a Chebyshev low pass filter with corner frequency of 10kHz. Use the information in the Bode plot to answer the following questions in the boxes provided.

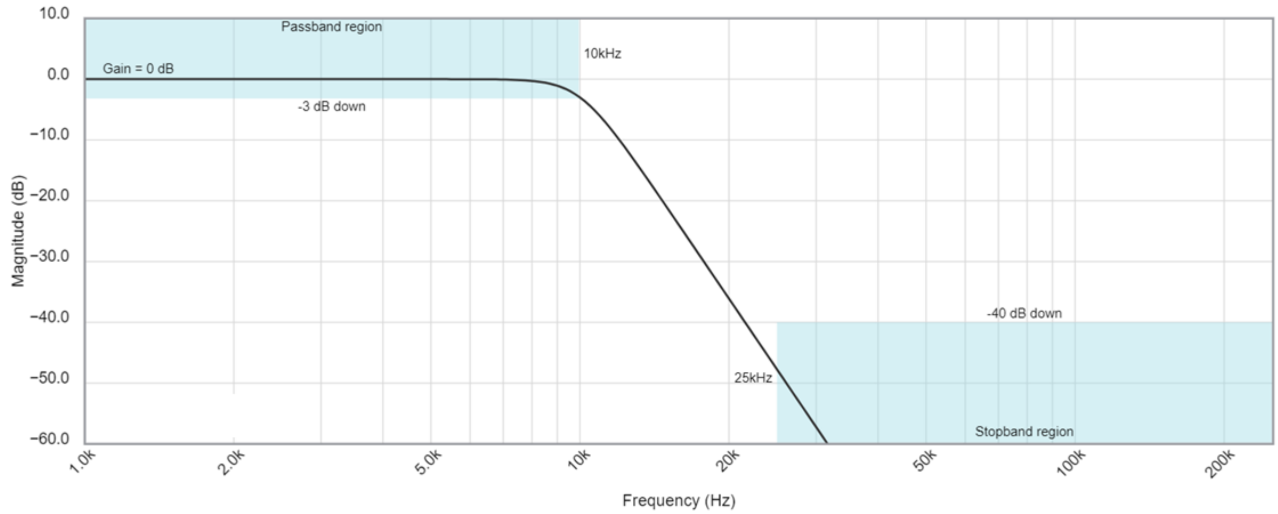


7. If you apply a 20kHz sin wave to this filter, how much attenuation will the output experience?

8. You want -45dB of attenuation, what frequency sin wave should you apply?

9. Estimate the slope of the magnitude in the stop band and use this to estimate the order of the filter. Hint, use the corner frequency as one of your points.

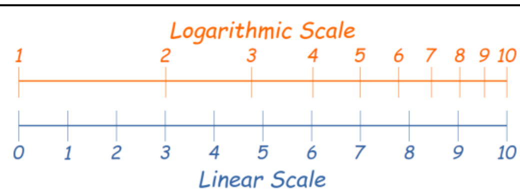
The following Bode plot is from a Butterworth low pass filter with corner frequency of 10kHz. Use the information in the Bode plot to answer the following questions in the boxes provided.



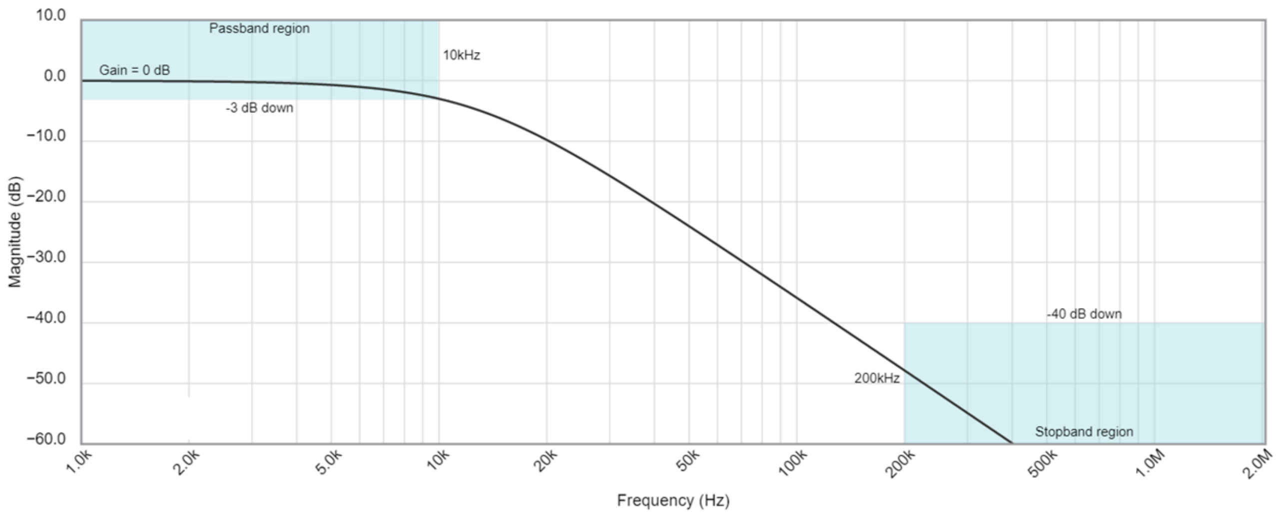
10. If you apply a 25kHz sin wave to this filter, how much attenuation will the output experience?

11. You want -35dB of attenuation, what frequency sin wave should you apply?

12. Estimate the slope of the magnitude in the stop band and use this to estimate the order of the filter. Hint, use the corner frequency as one of your points.



The following Bode plot is from a Bessel low pass filter with corner frequency of 10kHz. Use the information in the Bode plot to answer the following questions in the boxes provided.

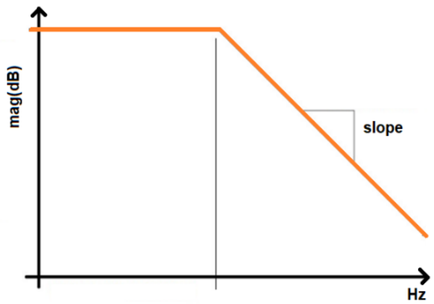


13. If you apply a 20kHz sin wave to this filter, how much attenuation will the output experience?

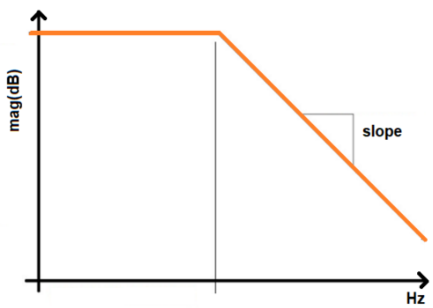
14. You want -55dB of attenuation, what frequency sin wave should you apply?

15. Estimate the slope of the magnitude in the stop band and use this to estimate the order of the filter. Hint, use the corner frequency as one of your points.

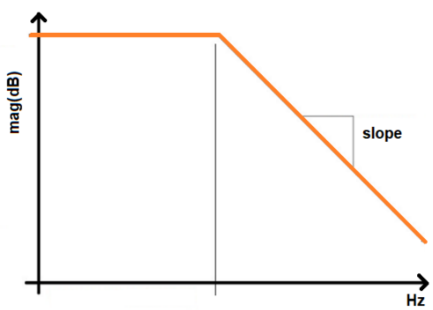
16. Determine the output attenuation (in dB) from a 80kHz input signal from a 1^{st} order LPF with a corner frequency of 2kHz that has 0dB of attenuation in the passband.



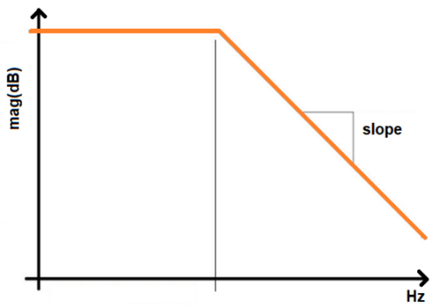
17. Determine the input frequency needed to attenuate an input signal by -48dB from a 1^{st} order LPF with a corner frequency of 2kHz that has 0dB of attenuation in the passband.



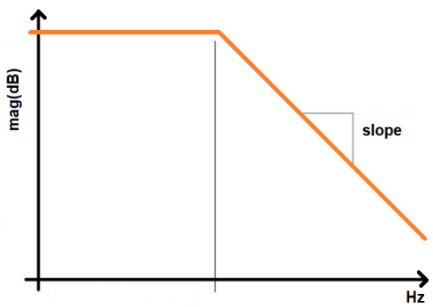
18. Determine the corner frequency of a 4^{th} order LPF that has 0dB of attenuation in the passband and attenuates a 130kHz input by -68dB.



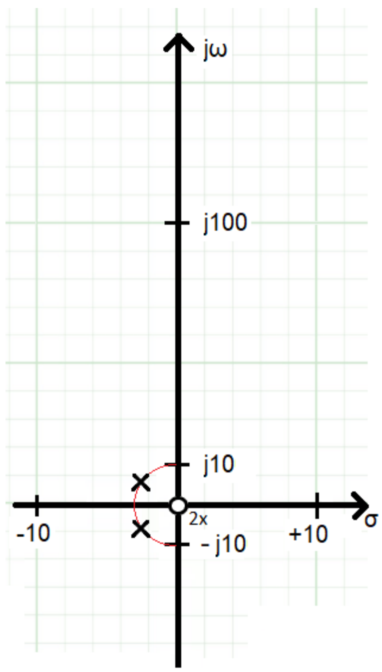
19. Determine the order of a filter that has 0dB of attenuation in the passband, a 15kHz corner frequency and attenuates a 90kHz input by -77.8dB.

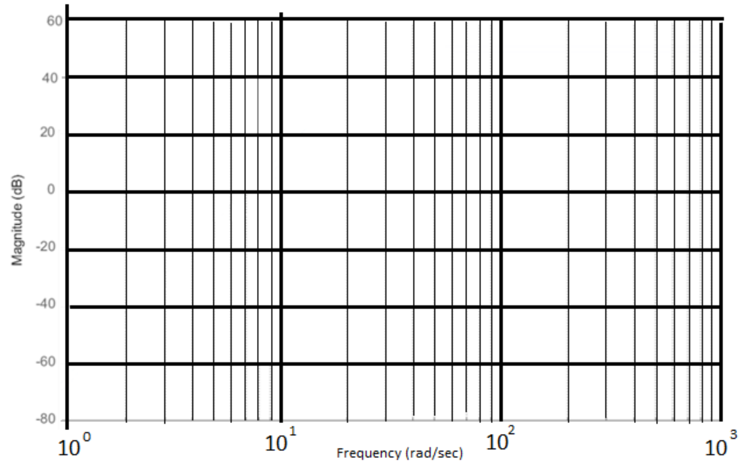


20. Determine the magnitude in the passband of a second order LPF, an 8kHz corner frequency and attenuates a 56kHz input by -13.8dB.

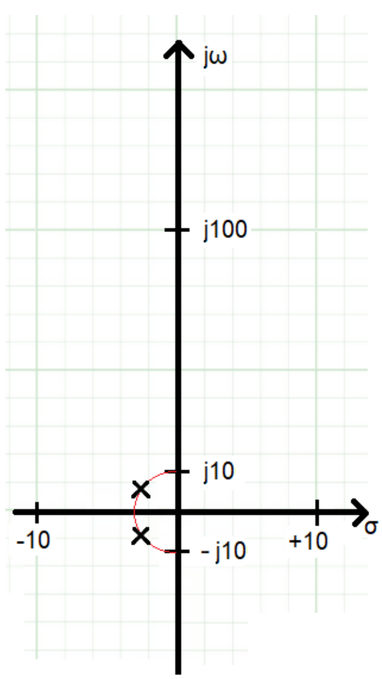


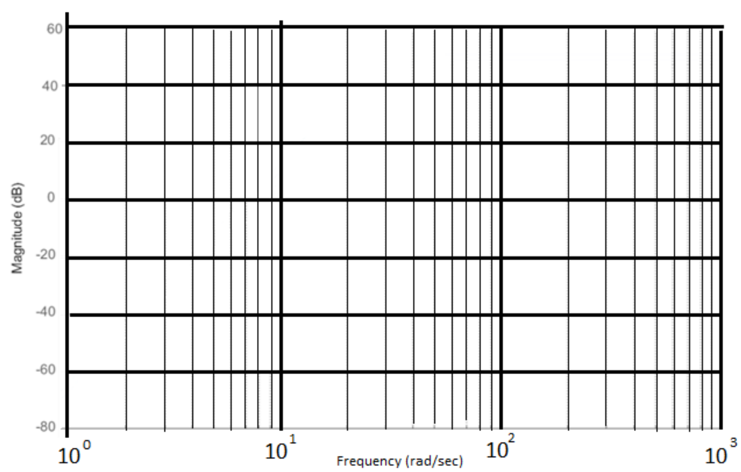
21. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently. There are two zeros at the origin.



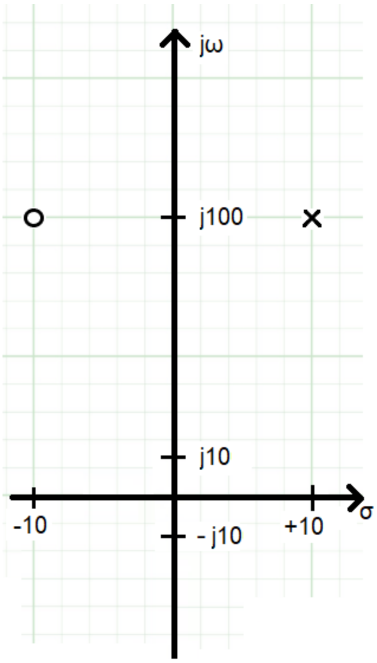


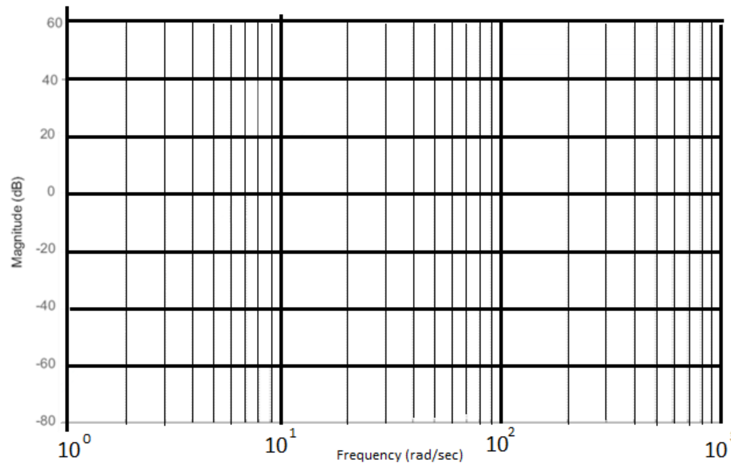
22. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently.



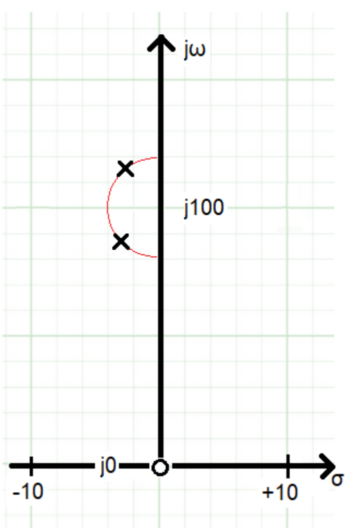


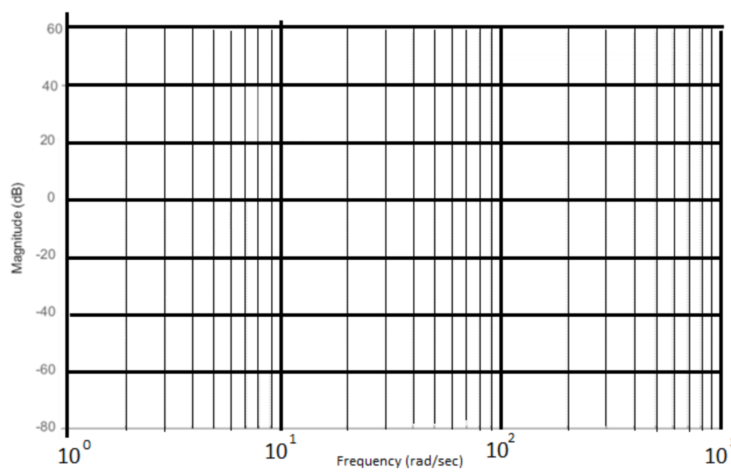
23. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently.



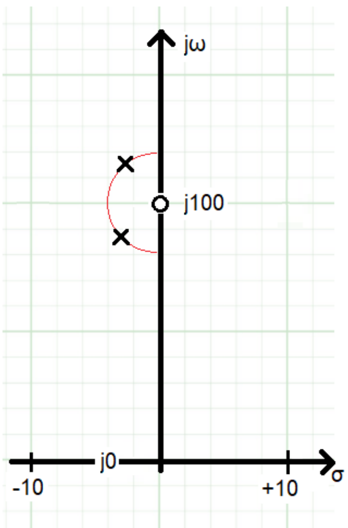


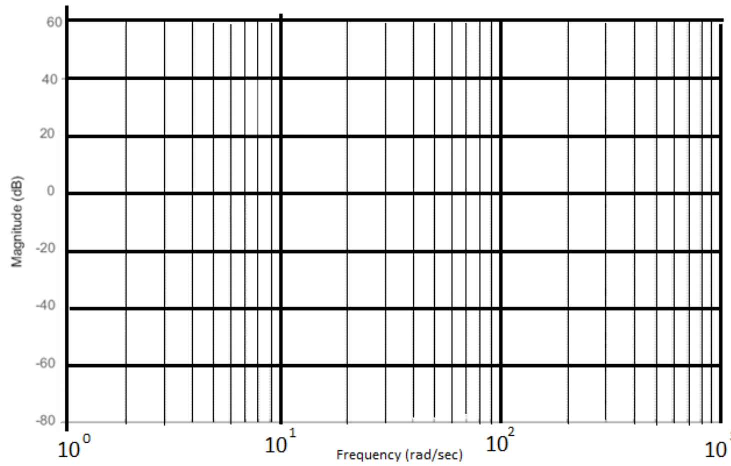
24. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently. There is one zero a the origin.





25. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently. There is one zero at j100.





26. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently.